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Edexcel GCSE Geography B (1GB0) — Component 3: People and Environment Issues · Topic 9: Consuming Energy Resources

3.3 Consuming Energy Resources

1. Types of Energy Resources

Type	Definition	Examples
Non-renewable	Finite resources that will run out because they cannot be replaced within a human lifetime	Coal, oil, natural gas, uranium (nuclear)
Renewable	Resources that are naturally replenished and will not run out	Solar, wind, HEP, tidal, wave, geothermal
Recyclable	Resources that can be reused or reprocessed	Biomass (if replanted), nuclear fuel reprocessing, energy from waste

2. Areas of Energy Surplus and Deficit

Energy security — having access to reliable, affordable and sufficient sources of energy. **Energy insecurity** — when supply is unreliable, too expensive or insufficient to meet demand.
Energy surplus — a country produces more energy than it uses (e.g. Saudi Arabia, Russia, Norway, Qatar). **Energy deficit** — a country uses more energy than it produces and must import the rest (e.g. Japan, South Korea, Germany, the UK, India).

- Energy resources are **unevenly distributed**: the Middle East holds around half of the world's proven oil reserves; Russia, Iran and Qatar hold most natural gas; China, the USA, India and Australia have the largest coal reserves.

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- Consumption is also uneven: the average person in the USA uses several times more energy than the global average and dozens of times more than someone in Ethiopia or Niger.

3. Reasons for Increasing Energy Consumption

- **Economic development** — as countries industrialise, factories, transport and infrastructure need more energy. China's energy use roughly tripled between 2000 and 2020.
- **Rising population** — more people need more energy for homes and transport.
- **Technology** — more devices (phones, computers, air-conditioning, electric vehicles, data centres) need more electricity.
- **Rising living standards** — wealthier people own more cars and appliances and travel more.

4. Factors Affecting Energy Supply

Factor	Effect
Physical factors	Fossil fuels are found only where the geology is right; renewable energy depends on geography — e.g. hydro needs high rainfall and steep valleys, geothermal needs tectonic activity, solar needs sunshine. Some reserves are in hostile environments (e.g. the Arctic, deep oceans).
Cost of exploitation and production	Deep-water oil, oil sands and Arctic drilling are expensive; when energy prices fall, these become uneconomic. Renewables have high start-up costs but low running costs.
Technology	New technology opens up previously inaccessible reserves (e.g. fracking for shale gas in the USA) and makes renewables cheaper and more efficient.
Political factors	Conflict and political instability disrupt supply — e.g. Russia's invasion of Ukraine (2022) caused gas prices in Europe to soar. Governments may restrict exploitation for environmental reasons (e.g. the UK fracking moratorium).

5. Environmental Impacts of Extracting Energy

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- **Coal mining:** opencast mines destroy landscapes and habitats; spoil heaps; methane release; burning coal produces the most CO_2 per unit of energy.
- **Oil and gas:** oil spills (Deepwater Horizon, Gulf of Mexico, 2010 — about 4.9 million barrels); pipelines through sensitive areas; gas flaring (Niger Delta).
- **HEP:** reservoirs flood valleys and forests, displace people and block fish migration.
- **Wind and solar:** land use, visual impact and mining of materials for turbines, panels and batteries.

6. Global Oil: Supply, Demand and Price

- **Supply:** the largest producers are the USA, Saudi Arabia and Russia. Over half of proven reserves are in the Middle East. OPEC (Organization of the Petroleum Exporting Countries) controls a large share of supply and can influence prices by changing production.
- **Demand:** the largest consumers are the USA, China and India. Demand is growing fastest in emerging economies as car ownership and industry grow.
- **Price:** oil prices are volatile — about $147 a barrel in July 2008, below $30 in early 2016, briefly negative in April 2020 (US futures, during the pandemic), and above $120 in 2022 after Russia invaded Ukraine. Prices rise when supply is disrupted (conflict, OPEC cuts) or demand grows, and fall during recessions.

7. Unconventional Fossil Fuels

Source	Advantages	Disadvantages
Oil sands (Alberta, Canada)	Vast reserves (among the largest in the world); jobs and revenue; secure supply for North America	Destroys boreal forest and peat; toxic tailings ponds; high water and energy use; higher emissions than conventional oil
Shale gas (fracking)	Made the USA a major gas exporter; gas emits less CO_2 than coal	Earth tremors, groundwater contamination risk, methane leaks, water use
Deep-water oil	Large untapped reserves (e.g. off Brazil)	Expensive and risky — Deepwater Horizon explosion killed 11 workers and caused a huge spill

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Arctic oil and gas	Large potential reserves as sea ice retreats	Fragile environment; spills very hard to clean up in ice; extreme conditions

8. Strategies to Increase Energy Supply

Renewable sources

Source	Advantages	Disadvantages
Wind	No emissions when running; costs have fallen sharply; offshore wind is very productive	Intermittent; visual and noise impact; can harm birds
Solar	Cheap and falling in cost; works at small scale in remote areas	Only generates in daylight; less effective in cloudy places; large solar farms use farmland
Hydroelectric (HEP)	Reliable and controllable; reservoirs can store water	Dams flood valleys, displace people and disrupt river ecosystems
Tidal and wave	Predictable tides; huge potential around the UK	Expensive; can affect estuary habitats; technology still developing
Geothermal	Reliable, low emissions	Only possible in tectonically active areas (e.g. Iceland, Kenya)
Biomass	Uses waste; can be "carbon neutral" if replanted	Crops compete with food for land; burning releases pollution

Non-renewable sources

- **Fossil fuels** (coal, oil, gas) — reliable and flexible, with existing infrastructure, but finite and the largest cause of greenhouse gas emissions and air pollution.
- **Nuclear power** — very low carbon and reliable, but expensive to build, produces radioactive waste and carries a small risk of serious accidents (Chernobyl 1986, Fukushima 2011).

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9. Reducing Reliance on Fossil Fuels

- **Energy efficiency and conservation** — insulation, LED lighting, efficient appliances and vehicles, smart grids.
- **Alternative energy** — expanding wind, solar, HEP and nuclear; electric vehicles; hydrogen for heavy industry and transport.
- **Carbon capture and storage** — trapping CO_2 from power stations and industry and storing it underground.
- **Government policy** — carbon taxes, subsidies for renewables, net-zero targets (the UK aims for net zero by 2050).

10. Moving Towards a Sustainable Resource Future

Individual energy use and carbon footprints

- A person's **carbon footprint** is the total greenhouse gas emissions caused by their activities — heating, travel, diet and the products they buy.
- Individuals can reduce this by insulating homes, using public transport, cycling or walking, eating less meat and buying fewer new goods.

Energy conservation

Area	Strategies
Designing homes and workplaces	Loft and cavity-wall insulation, double or triple glazing, solar panels, heat pumps, passive solar design (large south-facing windows), green roofs, smart meters. BedZED in London was designed as a zero-fossil-energy development.
Transport	Electric vehicles (new petrol and diesel car sales in the UK are due to end in 2030); integrated public transport; cycle lanes; congestion charges and low-emission zones; car-sharing.
Demand reduction	Switching off appliances; using energy-efficient appliances and LED lighting; smart meters that show energy use in real time.

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Use of technology	Improving the efficiency of power stations, vehicles and appliances; carbon capture and storage; smart grids that match supply and demand.

11. Attitudes to Energy Use

Stakeholder	Typical attitude
TNCs (e.g. oil companies)	Want to keep extracting fossil fuels profitably, though many now invest in renewables
Governments	Balance energy security, cost to consumers, jobs and climate targets
Climate scientists	Urge rapid cuts in fossil fuel use to limit warming to 1.5–2°C
Environmental groups (Greenpeace, Friends of the Earth)	Campaign against new fossil fuel extraction; support renewables and efficiency
Consumers	Want affordable, reliable energy; attitudes vary with income and awareness

Exam Tip: Component 3 questions often ask you to consider different stakeholders' views. Show that attitudes depend on who benefits and who pays — for example, an oil-producing country and a small island state threatened by sea level rise will see energy policy very differently.

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